3-27-26 (WEEK 5)

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DEFINITION

Say L/K: fin Gal, $G = \mathrm{Gal}(L/K)$.

For some max $P \trianglelefteq \mathcal{O}_K$, $P\mathcal{O}_L = \prod_{i=1}^{g} B_i^e$, g: # distinct max ideals lying over p

Then, the decomposition group of B_i is said to be $G_{B_i} := \{\sigma \in \mathrm{Gal}(L/K) \mid \sigma(B_i) = B_i\}$

Since $G \curvearrowright \{B_i\}$ transitively, $G_{B_i} = \sigma_i G_{B_1} \sigma_i^{-1}$ for some $\sigma_i \in G$

IDEAL NORMS AND DISCRIMINANT IDEAL

RECALL

For $\mathcal{O}_K = \mathbb{Z}$, $K = \mathbb{Q}$,

If $0 \neq a \trianglelefteq \mathcal{O}_K$, we def $N_{L/K}(a) = [\mathcal{O}_L : a]$ ($\mathcal{O}_L$: free $\mathbb{Z}$-mod of fin rank)

DEFINITIONS

Naïve mapping: $\mathrm{Spec}\,\mathcal{O}_L \longrightarrow \mathrm{Spec}\,\mathcal{O}_K$

$B \longmapsto B \cap \mathcal{O}_K$

Check: $\mathcal{O}_K = \mathbb{Z}$, $B \trianglelefteq \mathcal{O}_L$: $B \cap \mathbb{Z} = (p)$, p: prime by PID

Then, $[\mathcal{O}_L : B] = p^f$, $f = [\mathcal{O}_L/B : \underbrace{\mathcal{K}(p)}_{\mathbb{Z}/p}]$ NOT p.

Thus, we define $f(B|p) := [\mathcal{O}_L/B : \mathcal{K}(p)]$ and map $B \to (B \cap \mathcal{O}_K)^{f(B|p)}$

As we have unique factorization of ideals in $\mathcal{O}_L$, we can extend the N_r ($N_{L/K}$, but abbrev since no ambiguity) map multiplicatively to all nonzero ideals; and define $N_r(0) = 0$ and $N_{L/K}(\mathcal{O}_L) = \mathcal{O}_K$. We can also extend the $N_{L/K}$ map to all fractional ideals of $\mathcal{O}_L$ via $(B^{-1}) \longrightarrow N_r(B)^{-1}$ ($BB^{-1} = \mathcal{O}_L$)

OBSERVE

$\{\text{Ideals in } \mathcal{O}_K\} \hookrightarrow \{\text{Ideals in } \mathcal{O}_L\}$

$I \longrightarrow I\mathcal{O}_L$

Since $\mathcal{O}_K$: DDK, $(0) \neq I = \prod_{i=1}^{n} P_i^{e_i}$, P_i: max

For $P_i \neq P_j$, we know $\{B \mid B \supseteq P_i\mathcal{O}_L\} \cap \{B \mid B \supseteq P_j\mathcal{O}_L\} = \emptyset$, so $I\mathcal{O}_L = \prod_{i=1}^{n} \prod_{B \mid P_i} B^{e_i}$

For $(0) \neq I, J \trianglelefteq \mathcal{O}_K$, $J = \prod_{i=1}^{n} P_i^{\varepsilon_i}$, $\varepsilon_i, e_i \geq 0$, if $J\mathcal{O}_L = I\mathcal{O}_L$, $\varepsilon_i = e_i$, then $I = J$ □

REMARK

For max $P \trianglelefteq \mathcal{O}_K$, $N_{L/K}(P\mathcal{O}_L) = p^{[L:K]}$

$P\mathcal{O}_L = \prod_{i=1}^{n} B_i^{e_i}$, then $N_r(P\mathcal{O}_L) = \prod N_r(B_i)^{e_i} = \prod [\underbrace{(B_i \cap \mathcal{O}_K)}_{P}^{f_i}]^{e_i} = p^{\sum f_i e_i} = p^{[L:K]}$

LEMMA

Say L/K: fin Gal, $G = \mathrm{Gal}(L/K)$.

Let $B \trianglelefteq \mathcal{O}_L$ be max and $P = \mathcal{O}_K \cap B$ ($P\mathcal{O}_L = \prod_{i=1}^{n} B_i^e$, $B_1 = B$)

Then, $N_{L/K}(B)\mathcal{O}_L = (\prod_{i=1}^{n} B_i)^{ef} = \prod_{\sigma \in G} \sigma(B)$

<u>Proof</u>

By cro, $[L:K] = nfg$

Then, $N_{L/K}(B)\mathcal{O}_L = (p^f)\mathcal{O}_L = (p\mathcal{O}_L)^f = (\prod_{i=1}^{n} B_i)^{ef}$, $n = \frac{|G|}{fg}$

$= (\prod_{\sigma \in G/G_B} \sigma(B_1))^{ef} = \prod_{\sigma \in G} \sigma(B_1)$ □

($G/G_B \overset{1-1}{\longleftrightarrow} \{B_1, \dots, B_n\} \Rightarrow |G_B| = fg$)

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PROPOSITION

Let L/K: fin+sep for $x \in \mathcal{O}_L$. Then, $Nr_{L/K}(x\mathcal{O}_L) = Nr_{L/K}(x)\mathcal{O}_L$

Proof

We only need to check $Nr(x\mathcal{O}_L)\mathcal{O}_L = N_{L/K}(x)\mathcal{O}_L$

[1] $x=0$ or $x \in \mathcal{O}_L^\times$: Obvious

[2] $x \neq 0$ and $x \notin \mathcal{O}_L^\times$: $x\mathcal{O}_K \trianglelefteq \mathcal{O}_L$: nonzero proper ideal

$x\mathcal{O}_L = \prod_{i=1}^n B_i^{e_i}$, $B_i \trianglelefteq \mathcal{O}_L$: distinct max

① Special case: L/K: Gal, $G = Gal(L/K)$

Then, $Nr_{L/K}(x)\mathcal{O}_L = \prod_{\sigma \in G} \sigma(x)\mathcal{O}_L = \prod_{\sigma \in G}\prod_{i=1}^n \sigma(B_i)^{e_i} = \prod_{i=1}^n \prod_{\sigma \in G} \sigma(B_i)^{e_i} = \prod_{i=1}^n Nr(B_i)^{e_i}\mathcal{O}_L = Nr(x\mathcal{O}_L)\mathcal{O}_L$

② L/K: Choose a Galois closure $\tilde{L}$ of L, s.t. $K \subseteq L \subseteq \tilde{L}$

Then, $Nr_{\tilde{L}/K}(x\mathcal{O}_{\tilde{L}}) = Nr_{L/K}(x\mathcal{O}_L)^{[\tilde{L}:L]} = Nr_{L/K}(x)^{[\tilde{L}:L]}\mathcal{O}_L$

$\therefore$ ① $\Rightarrow Nr_{L/K}(x\mathcal{O}_L)^{[\tilde{L}:L]} = N_{L/K}(x)^{[\tilde{L}:L]}\mathcal{O}_K$

By unique factorization of ideals, $N_{L/K}(x\mathcal{O}_L) = N_{L/K}(x)\mathcal{O}_L$ □

DISCRIMINANT IDEAL

DEFINITION

Let $J \subseteq L$ be an $\mathcal{O}_K$-mod containing a basis of L over K (☆)

We define the discriminant of J over $\mathcal{O}_K$, denoted by $\delta(J/\mathcal{O}_K)$ (or $\delta_{J/\mathcal{O}_K}$), as an $\mathcal{O}_K$-module generated by $\{disc(\underbrace{Tr(\alpha_i\alpha_j)_{i,j}}_{\text{choose basis } (\alpha_1, \alpha_2, \dots, \alpha_n)}) \mid \alpha_i \in J\} \subseteq K$

In particular, if $(0) \neq I \trianglelefteq \mathcal{O}_L$, then it satisfies (☆)

$\mathcal{O}_L$: contains a basis of L/K $\{\alpha_i\}$, then $\forall 0 \neq x \in I$, $\{x\alpha_i\}$ is a basis of L/K.

REMARK

J is a fractional ideal in $L \Rightarrow \exists a \in K^\times$, s.t. $aJ \trianglelefteq \mathcal{O}_L$

($b \in L^\times$, $bJ \subseteq \mathcal{O}_L \Rightarrow$ Take $a = Nr_{L/K}(b)$)

Then, $\underbrace{disc(aJ/\mathcal{O}_K)}_{\in \mathcal{O}_K} \supseteq \underbrace{a^{2n}}_{n=[L:K]}\delta(J/\mathcal{O}_K) \Rightarrow \delta(J/\mathcal{O}_K)$: fractional ideal in K

LEMMA

Say $S \subseteq \mathcal{O}_K \setminus \{0\}$ is a multiplicative set.

Let $(0) \neq J \trianglelefteq \mathcal{O}_L$, $S^{-1}\delta(J/\mathcal{O}_K) = \delta(S^{-1}J/S^{-1}\mathcal{O}_K)$

Proof (Sketch)

It is easy to check.

Let $S = \mathcal{O}_K \setminus P$, P: max ideal

$S^{-1}\mathcal{O}_L / S^{-1}\mathcal{O}_K$ ← DDK w/ only one max ⇒ principal

↳ only max ideal are those lying over P

$\therefore$ By exercise, $S^{-1}\mathcal{O}_L$: PID □

PROPOSITION

Say $(0) \neq I \trianglelefteq \mathcal{O}_L$. Then, $\delta(I/\mathcal{O}_K) = Nr_{L/K}(I)^2\delta(\mathcal{O}_L/\mathcal{O}_K)$

Proof

For each $p \trianglelefteq \mathcal{O}_K$, we check the p-powers appearing on both sides are the same.

In other words, check: $S^{-1}\delta(I/\mathcal{O}_K) = S^{-1}Nr(I)^2\delta(\mathcal{O}_L/\mathcal{O}_K) \overset{Lemma}{\Longrightarrow} \delta(S^{-1}I/\mathcal{O}_{K,P}) = Nr(I)^2\delta(S^{-1}\mathcal{O}_L/S^{-1}\mathcal{O}_K)$

Now, we can assume $\mathcal{O}_K, \mathcal{O}_L$ are PID. In particular, we need to check $I = (x)$, $x \in \mathcal{O}_L$ and $\mathcal{O}_L = \bigoplus_{i=1}^n \mathcal{O}_K\alpha_i \Rightarrow I = \bigoplus_{i=1}^n \mathcal{O}_K(x\alpha_i)$

Then, $\delta(I/\mathcal{O}_K) = (\det(Tr(x\alpha_i, x\alpha_j))_{i,j}) = \det(m_x)^2 \cdot \det(Tr(\alpha_i\alpha_j)_{i,j}) = Nr(x)^2\delta(\mathcal{O}_L/\mathcal{O}_K) \overset{\text{Prop}}{=} Nr(I)^2\delta(\mathcal{O}_L/\mathcal{O}_K)$ □

THEOREM (DEDEKIND-KUMMER)

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Fix a max ideal $p \trianglelefteq \mathcal{O}_K$.

Since L/K: fin + sep, we choose $\alpha \in \mathcal{O}_L$, s.t. $L = K(\alpha)$

Let $\Phi(x)$ = min poly of $\alpha \in \mathcal{O}_K[x]$

Say $\Phi(x) = \prod_{i=1}^{n} \varphi_i(x)^{e_i} \pmod p$, $\varphi_i(x)^{e_i} \pmod p$: distinct irr factor in $\mathcal{K}(p)[x]$, $\varphi_i(x) \in \mathcal{O}_K[x]$

Suppose $p \nmid \delta(\mathcal{O}_K(\alpha)/\mathcal{O}_K) \cdot \delta^{-1}(\mathcal{O}_L/\mathcal{O}_K)$. Then, $P\mathcal{O}_L = \prod_{i=1}^{n} B_i^{e_i}$, $B_i := (p, \varphi_i(\alpha))$